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APPLICATION EXAMPLE

PROFINET S2 system redundancy with SINAMICS G220 and SIMATIC S7-1500H

SINAMICS G220 / SIMATIC S7-1500 / TIA Portal V19

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1. Introduction

1.1. Overview

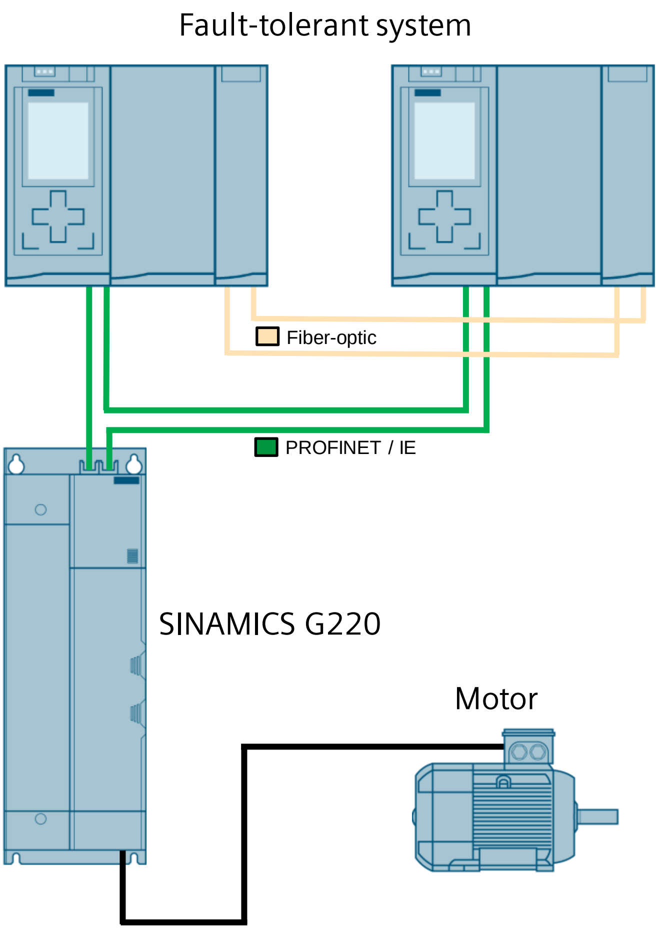
The drives of the SINAMICS G220 family support PROFINET S2 system redundancy. Thanks to S2 system redundancy, operation of the system is guaranteed even in the event of a defect or replacement of a controller, thus reducing downtimes.
The prerequisite for S2 system redundancy is a so-called S7-1500 fault-tolerant system, also referred to as an H system. An S7-1500 fault-tolerant system consists of 2 fault-tolerant controllers - the primary and backup CPU - that synchronize via fiber optics. If one controller fails, the other automatically takes over.

1.2. Principle of operation

S2 system redundancy

With system redundancy, a PROFINET device establishes several communication relationships to a redundant system. This example describes S2 system redundancy, in which a SINAMICS G220 is operated as a PROFINET device on a fault-tolerant system without the use of additional hardware.
The prerequisite for implementing system redundancy is the use of an S7-1500 fault-tolerant system. An S7-1500 fault-tolerant system consists of two high-availability (fault-tolerant) controllers (primary and secondary CPU). If one of the H CPUs fails, the other automatically takes over.

Schematic diagram



Advantages

- No plant downtime in the event of a controller failure
- Hot-swappable components during operation
- Changes to the configuration during operation
- Automatic synchronization after replacement of components

Limitations of SINAMICS

- PROFINET-IRT is not supported
- No simultaneous operation of Shared Device and system redundancy
- Maximum of 2 cyclic PROFINET connections
- System redundancy only via the PROFINET X150 interface
- While switching from one controller to the other, the setpoints of the last connection remain frozen and valid.

Limitations on the engineering system

Startdrive does not yet support the direct configuration of the drive as a client on a redundant system. Therefore, the SINAMICS G220 must be integrated into the redundant system via a GSDML file in the project.

Limitations of the S7-1500 fault-tolerant system

Limitations of the S7-1500 fault-tolerant System can be found in the system manual "SIMATIC S7-1500 Redundant System S7-1500R/H" (\3\).

Required knowledge

Basic knowledge of TIA Portal and SINAMICS G220 is required.

1.3. Components used

The following hardware and software components were used to create this application example:

Component	Quantity	Item number	Note
CPU S7-1517H System Bundle	1	6ES7 517-3HP00-0AB0	With Sync Modules
SINAMICS G220	1	6SL4112-0DA11-0AA0	Firmware V6.2
STEP 7 Professional V19	1	6ES7822-1AE23-0YA5	Engineering system in TIA Portal
Startdrive V19	1	6SL3072-4KA02-0XG5	Engineering software for SINAMICS drives in TIA Portal

You can obtain the listed components from the Siemens Industry Mall, for example.

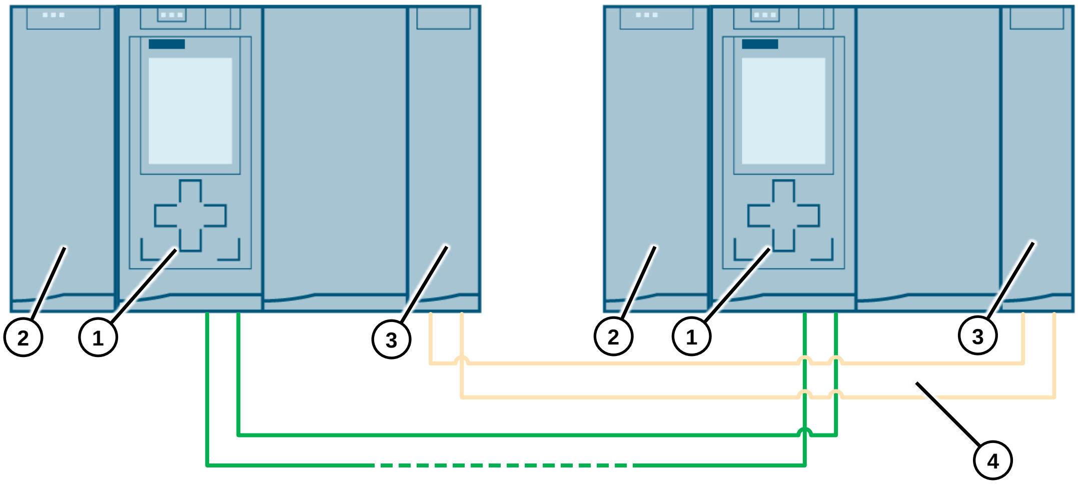
This application example consists of the following components:

Component	File name	Note
Documentation	109926733_S2_Systemredundancy_ S7-1500RH_G220_DOC_V10_en.pdf	This document
Example project	109926733_S2_Systemredundancy_ S7-1500RH_G220_PROJ_V10.zip	Example project for TIA Portal V19

2. Engineering

2.1. Hardware setup

The following images show the hardware setup and the wiring of the application:



S7-1500H (1)

The S7-1500H redundant system should be mounted either on one shared rail or on two separate mounting rails. You will connect the two CPUs with fiber-optic cables to two synchronization modules in each CPU. Set up the PROFINET ring via the CPUs' X1 P1 R and X1 P2 R PROFINET interface.

Power supply (2)

The peripheral power supply (PM) supplies the system power supply (PS) and core modules (CPU) with DC 24 V. For peripheral power supplies, we recommend the devices of the SIMATIC series. These devices can be mounted on the mounting rail.

Synchronization modules (S7-1500H only) (3)

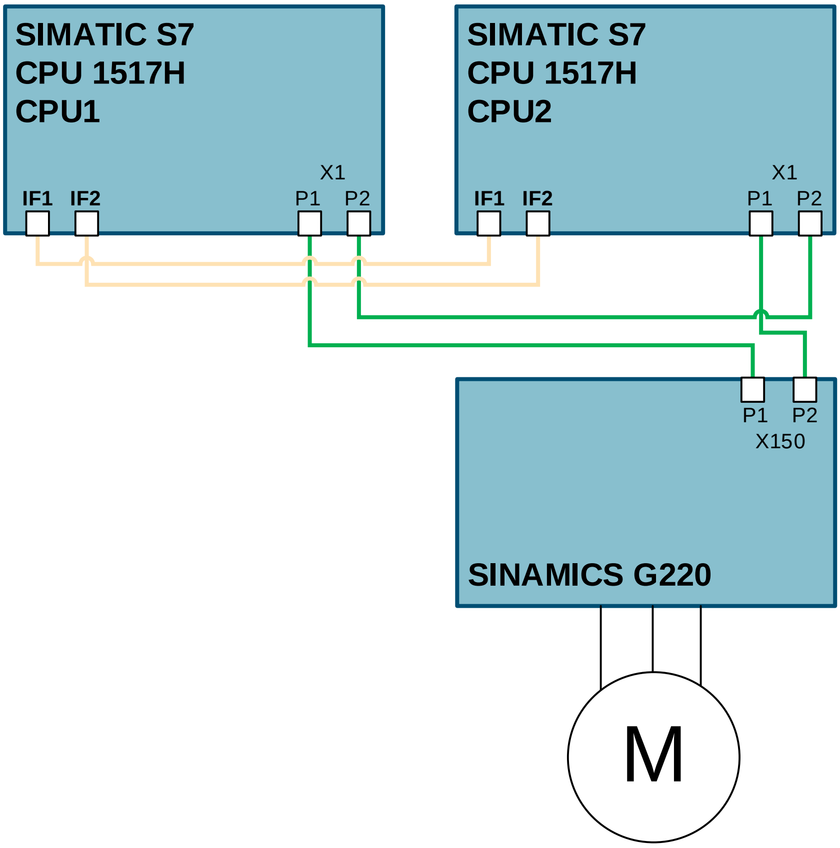
You will create two redundancy connections between the H CPUs with fiber-optic cables using a total of four synchronization modules (two in each H CPU).

Fiber-optic cables (S7-1500H only) (4)

You will connect the two synchronization modules for each CPU in a pair with fiber-optic cables.

2.2. Topology

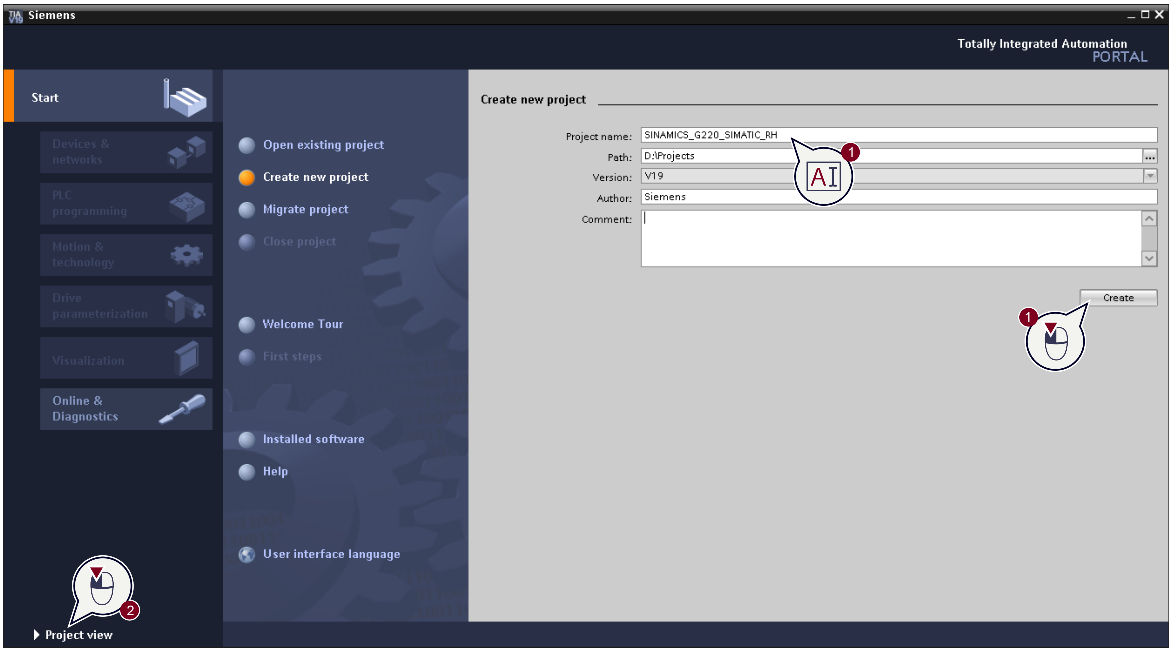
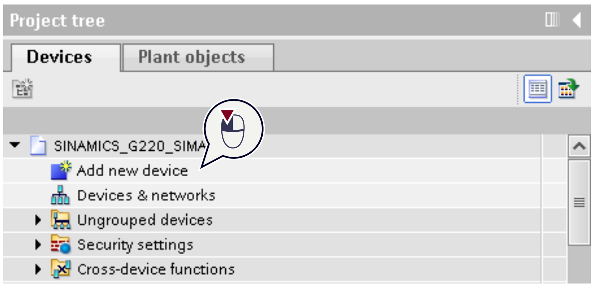
In order to integrate the SINAMICS G220 into the system via PROFINET, the following topology is used in the application example:



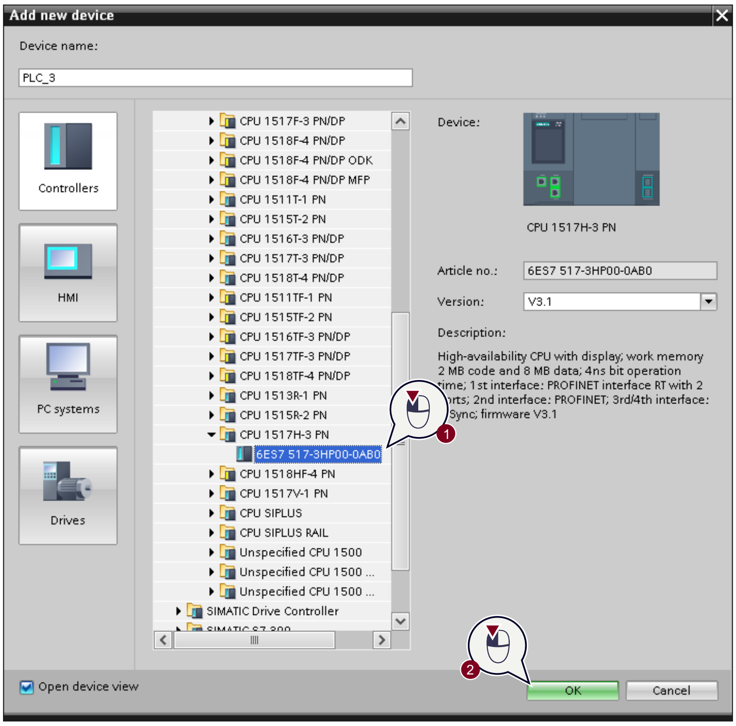
2.3. Configuration

2.3.1. Configuration of the SIMATIC H system

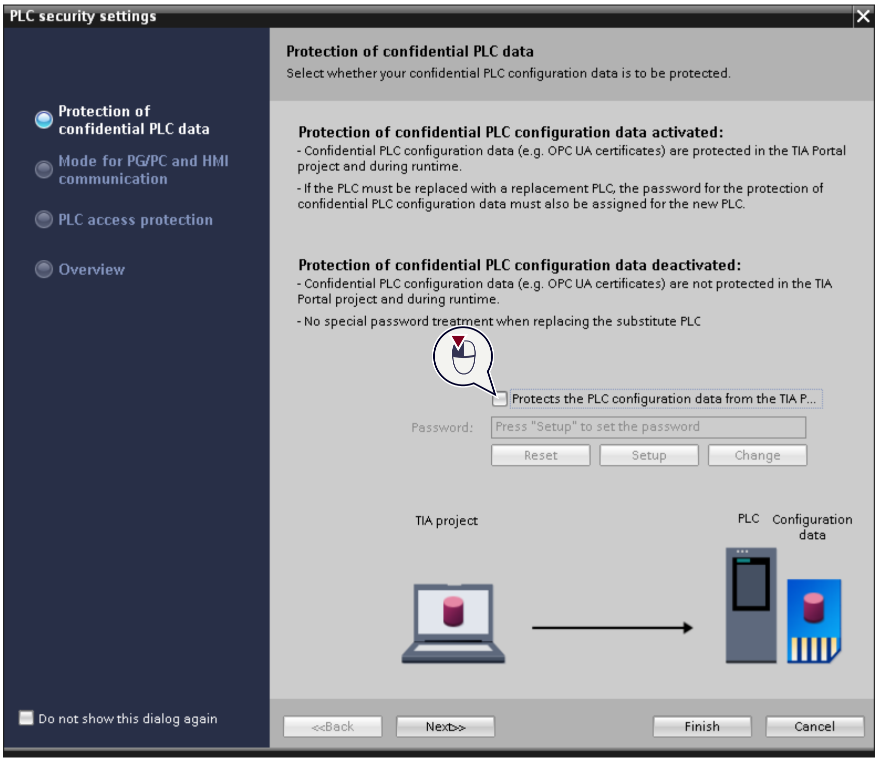
Follow the steps listed in the following table to configure the SIMATIC S7-1500 H system in TIA Portal:

No.	Action
1.	Create a new TIA Portal project (1) and switch to the project view (2). 
2.	Click on "Add new device" to create a new device in the project. 

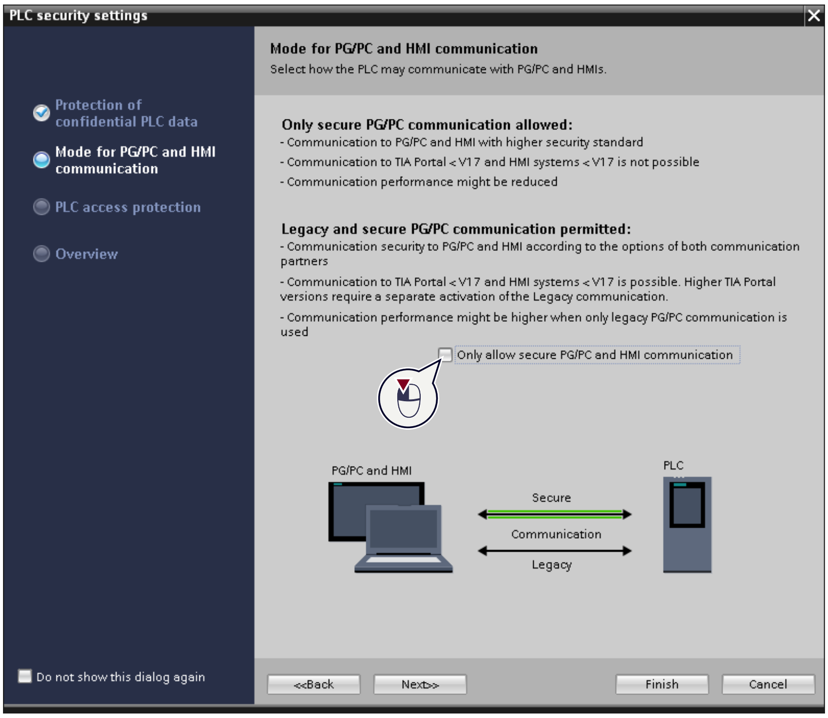
No.	Action
3.	In the dialog box, select the SIMATIC S7 H controller you are using (1) and confirm the selection with "OK" (2).



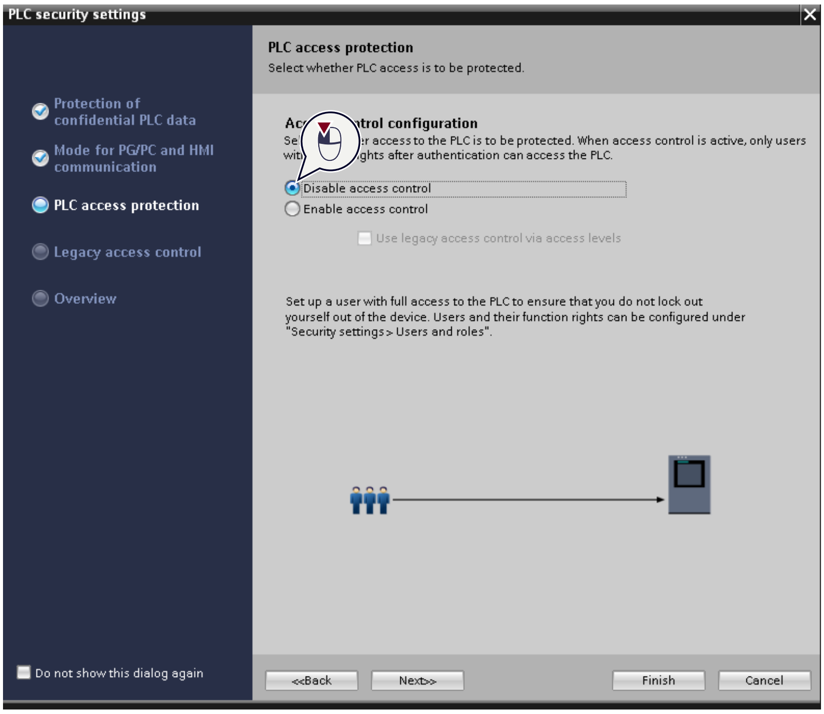
No.	Action
4.	**Note:** When a SIMATIC controller is created, the wizard for the controller's security settings is automatically opened. For the example project, no controller security functions will be configured in the next 3 steps. A machine in your project may require you to configure the security settings for the SIMATIC controller. Information on this can be found in the manual "SIMATIC S7-1500/ET 200MP Automation system". (\4\) For the sample project, disable the protection of confidential PLC data.



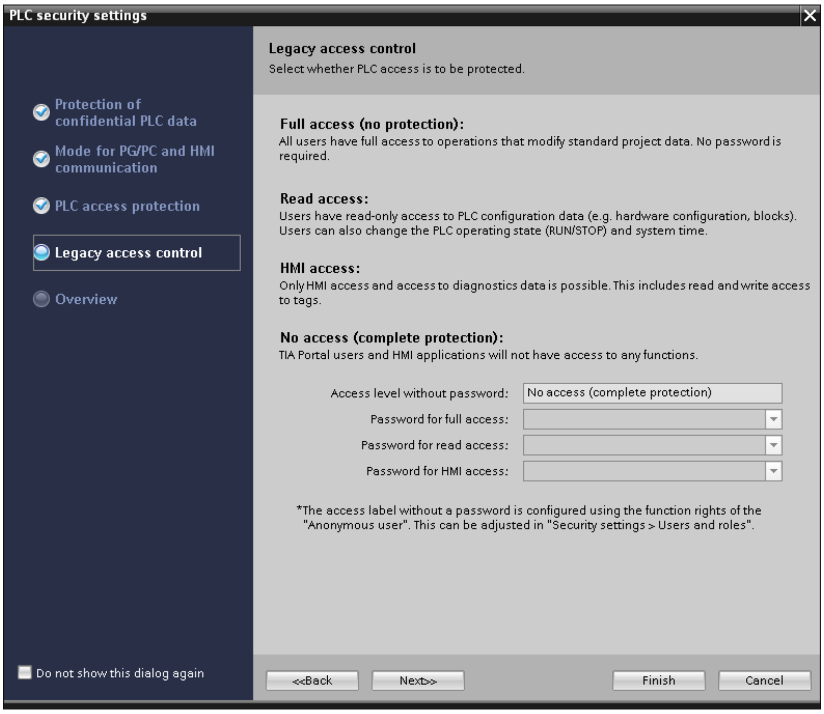
No.	Action
5.	Also enable "legacy" communication on the controller



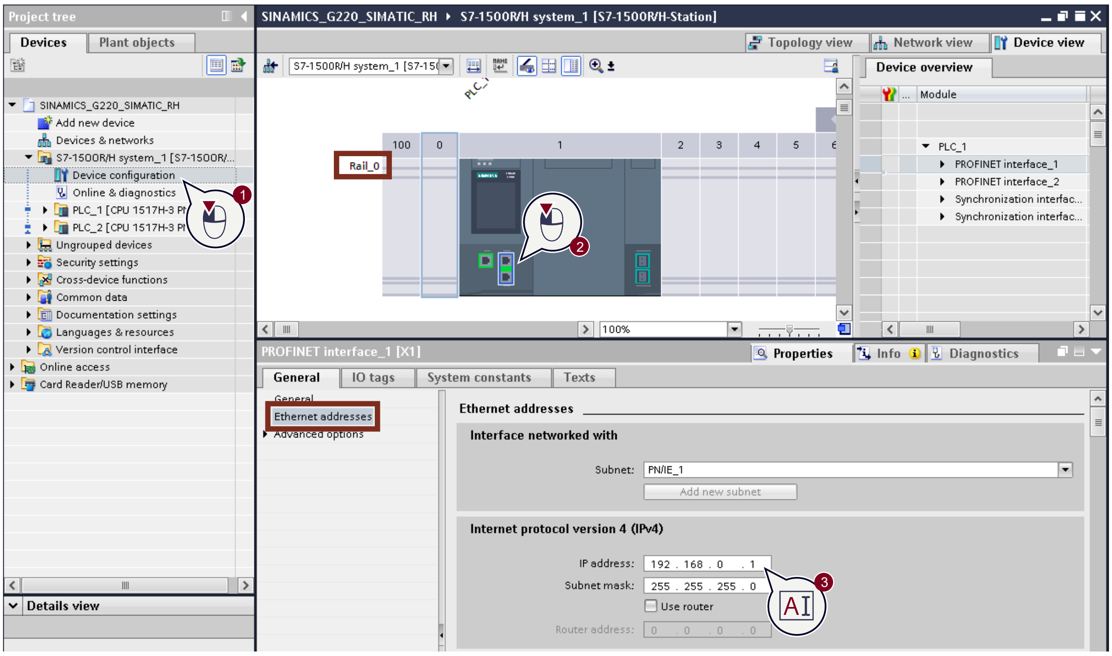
No.	Action
6.	Disable UMAC (User Management and Access Control) on the controller.



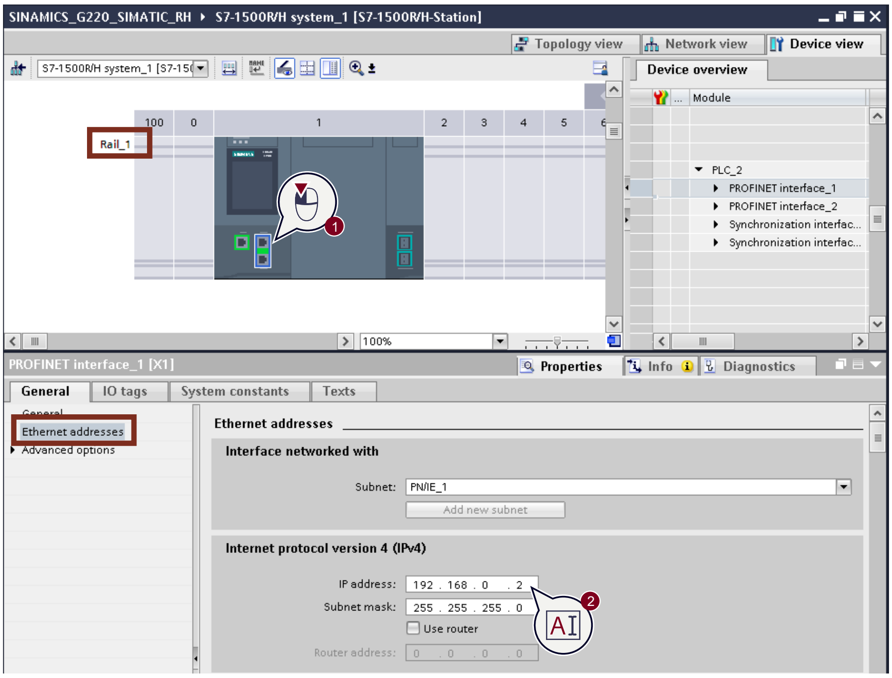
No.	Action
7.	Confirm the dialog by clicking on "Finish".



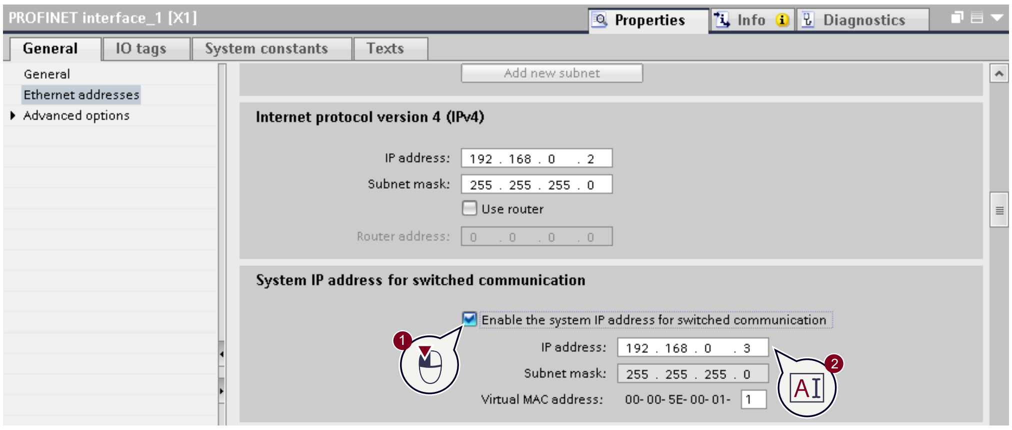
No.	Action
8.	If necessary, assign the IP address for the first controller of the S7-1500 H system by opening the device configuration (1), selecting the X1 PROFINET interface of the first SIMATIC S7 controller (2) and setting the desired IP address in the properties under "Ethernet addresses" (3). The example project keeps the default address of 192.168.0.1.



No.	Action
9.	If necessary, assign the IP address for the second controller of the S7-1500 H system by selecting the X1 PROFINET interface of the second SIMATIC S7 controller (1) and setting the desired IP address in the properties under "Ethernet addresses" (2). For the sample project, the default address is 192.168.0.2.



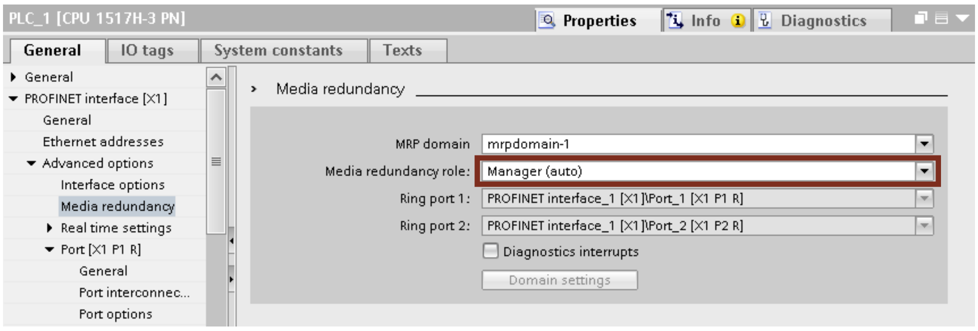
No.	Action
10.	**Recommended:** Optionally, activate and assign the system IP address for the X1 and X2 interfaces of the controllers by activating the option "Enable the system IP address for switched communication" (1) and adjusting the address if necessary (2).



Advantages over device IP addresses:

- The communication partner always communicates with the primary CPU of the S7-1500 fault-tolerant system via the same IP, regardless of which CPU is currently the primary CPU.
- The communication of the redundant S7-1500 H system via a system IP address continues to work when the primary CPU is changed.

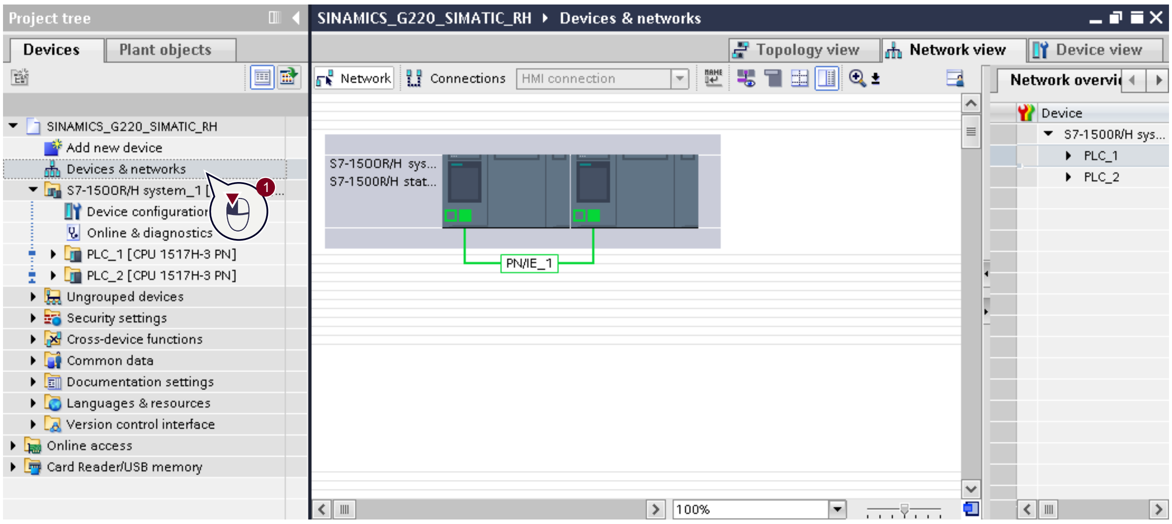
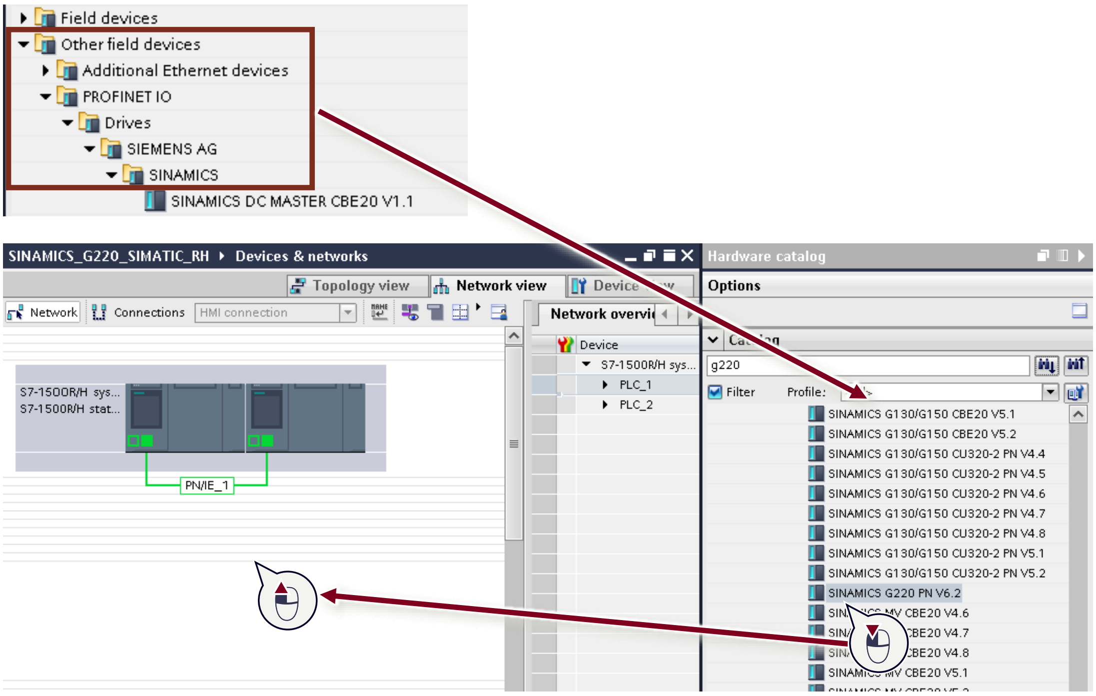
No.	Action
11.	Configure the S7-1500 H system as "Manager (auto)" in "Advanced options" > "Media redundancy".



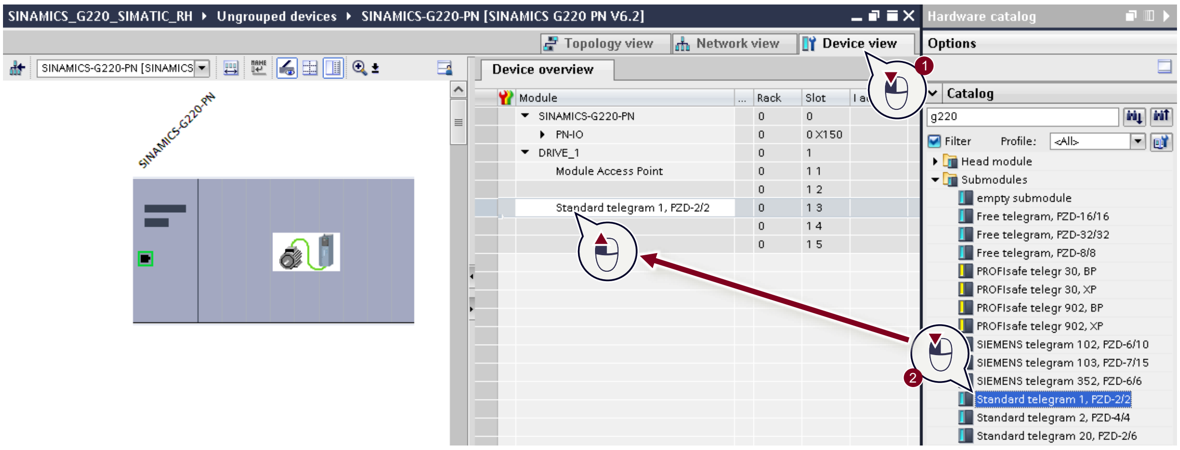
2.3.2. Configuration of the SINAMICS G220 drive

The SINAMICS G220 drive cannot be integrated directly into the redundant system with Startdrive. Instead, a GSDML file is used to connect the drive to the redundant system. This also makes it possible to configure the drive via the web server or in a separate TIA Portal project.

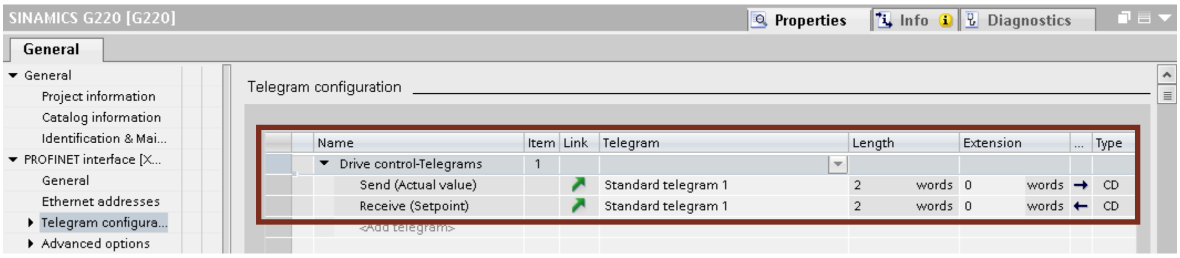
This example assumes basic configuration of the SINAMICS G220 drive is already complete, and the following table only explains the additional steps that are necessary to integrate the SINAMICS G220 into the redundant system.

No.	Action
1.	Open the Network view in the hardware configuration 
2.	Drag the GSDML for the SINAMICS G220 with the corresponding firmware version from the hardware catalog (Other field devices -> PROFINET IO -> Drives -> SIEMENS AG -> SINAMICS) into the hardware configuration. 

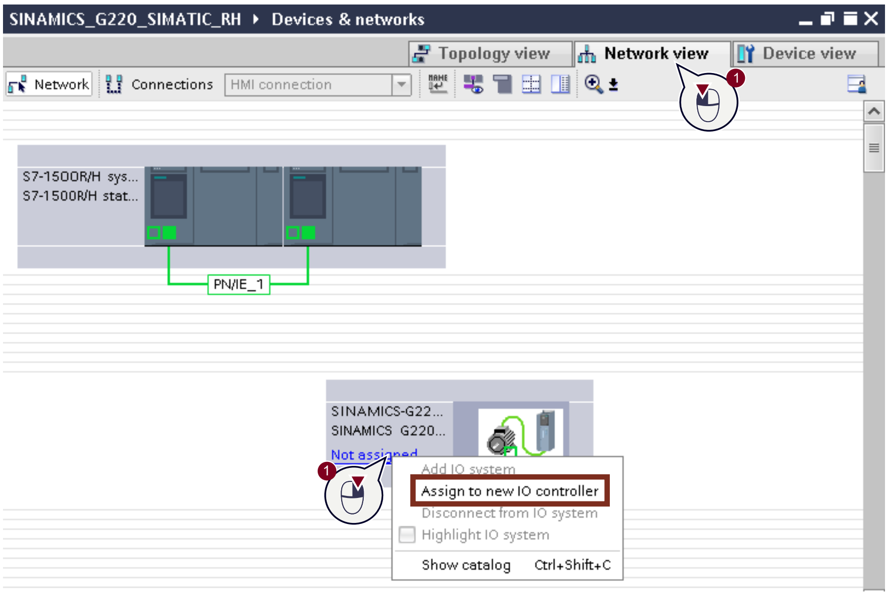
No.	Action
3.	Open the device view of the newly created SINAMICS G220 (1) via GSDML and drag and drop the used submodules (telegrams) into the device overview of the GSDML (2).

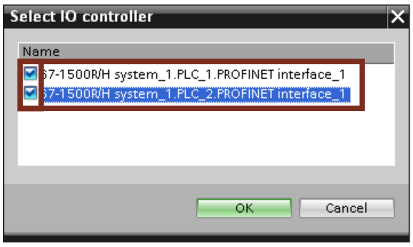
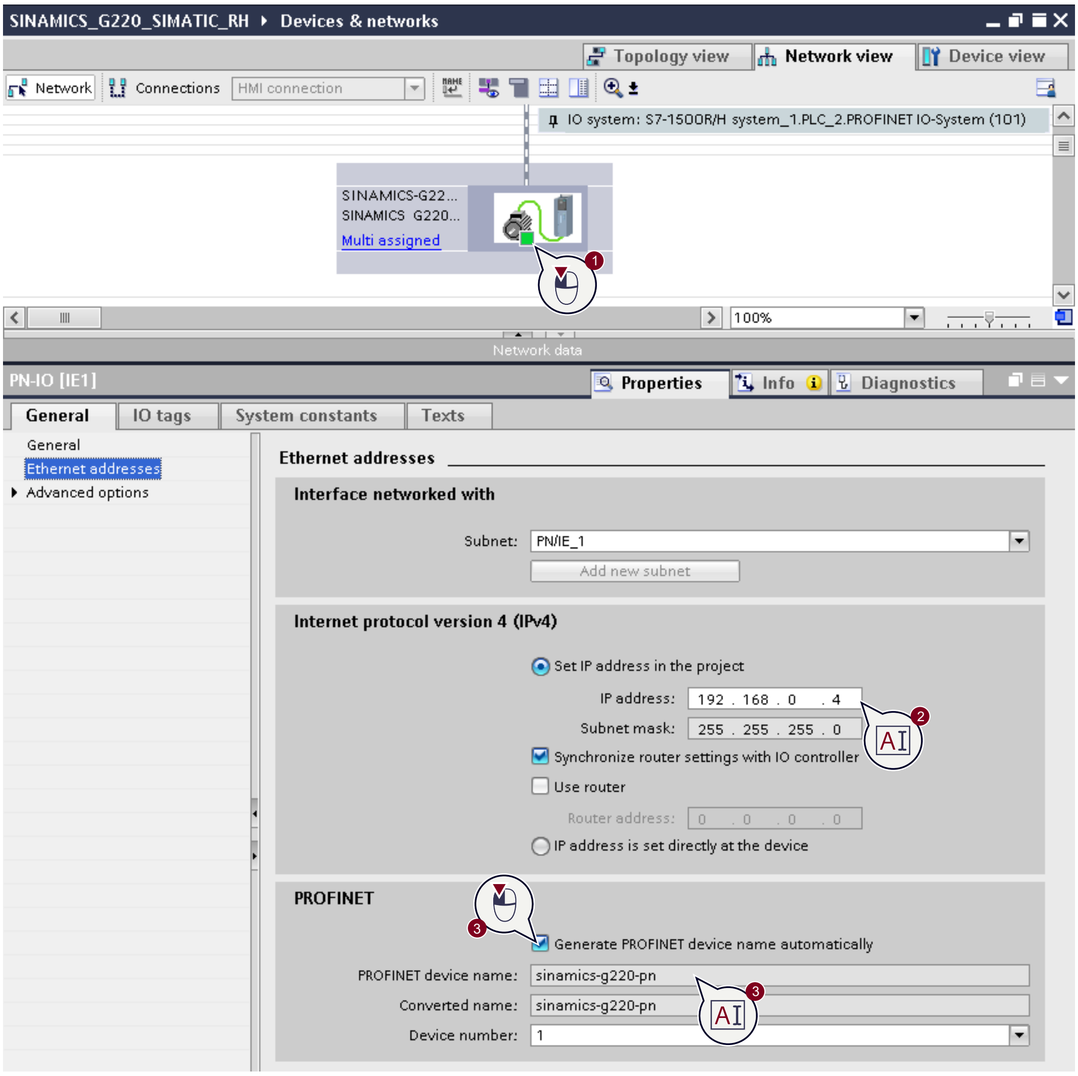


Note:
You can find the telegrams used by the drive in the telegram configuration within the Startdrive configuration:



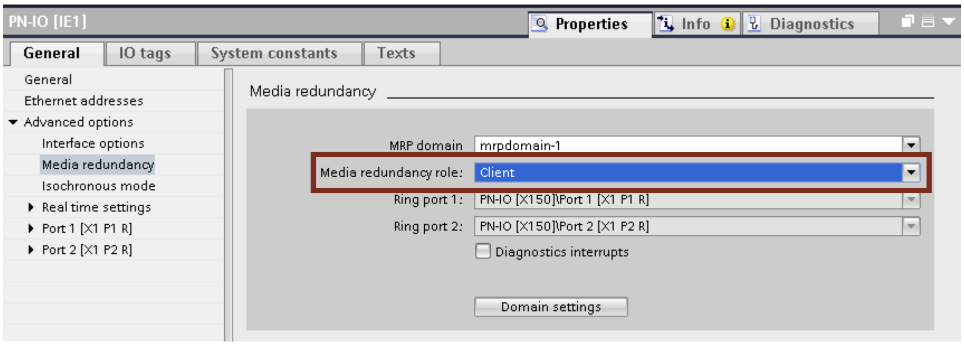
No.	Action
4.	Switch back to the Network view (1), right-click on "Not assigned" on the GSDML of the SINAMICS G220 and select "Assign to new IO controller".



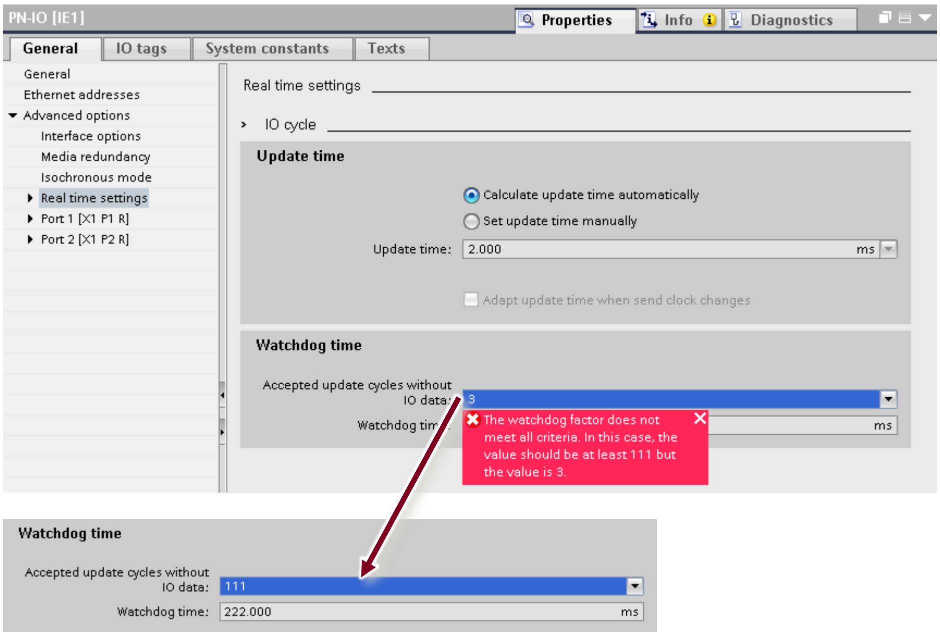
No.	Action
5.	In the dialog, select both controllers of the S7-1500 fault-tolerant system (check for both CPUs). 
6.	Select the PROFINET interface of the SINAMICS G220 (1) and, if necessary, assign the IP address (2) and the PROFINET device name (3). The IP address and the PROFINET device name of the GSDML must match the values in the drive configuration. The example uses the default values "192.168.0.4" and "sinamics-g220-pn". 

No.	Action

7. Configure the SINAMICS G220 as a "Client" in the advanced settings of the GSDML under Media Redundancy



8. In the real-time settings, set the watchdog factor to the value calculated by TIA Portal. It is shown in the tooltip.



3. Example project

3.1. Commissioning

If you use the components listed in chapter 1.3 and you have set up and networked the hardware as described in chapter 2.1 and 2.2, you can put the sample project into operation.

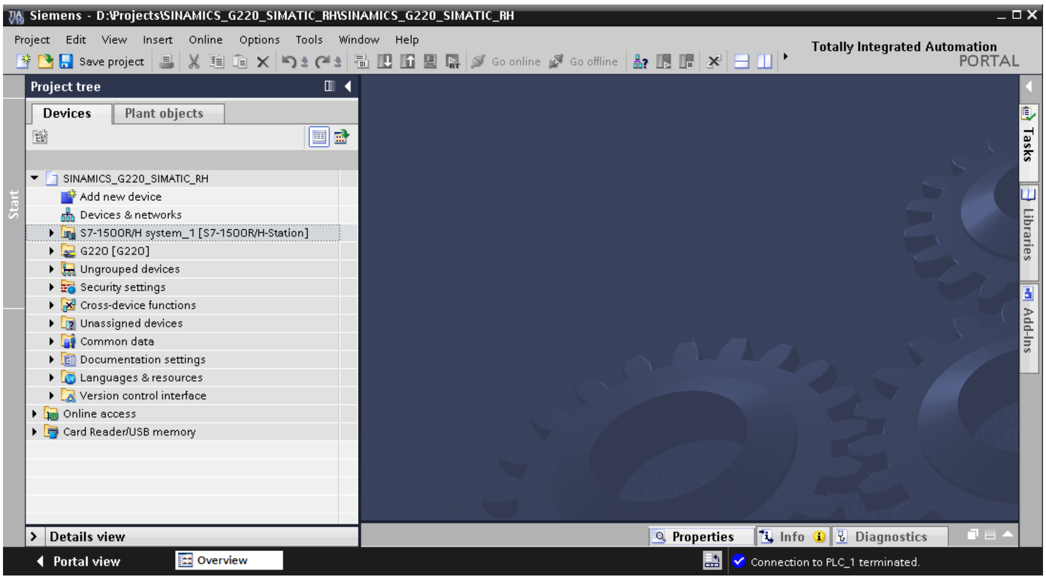
NOTE

In the hardware configuration of the sample project, the SINAMCIS G220 is included twice:

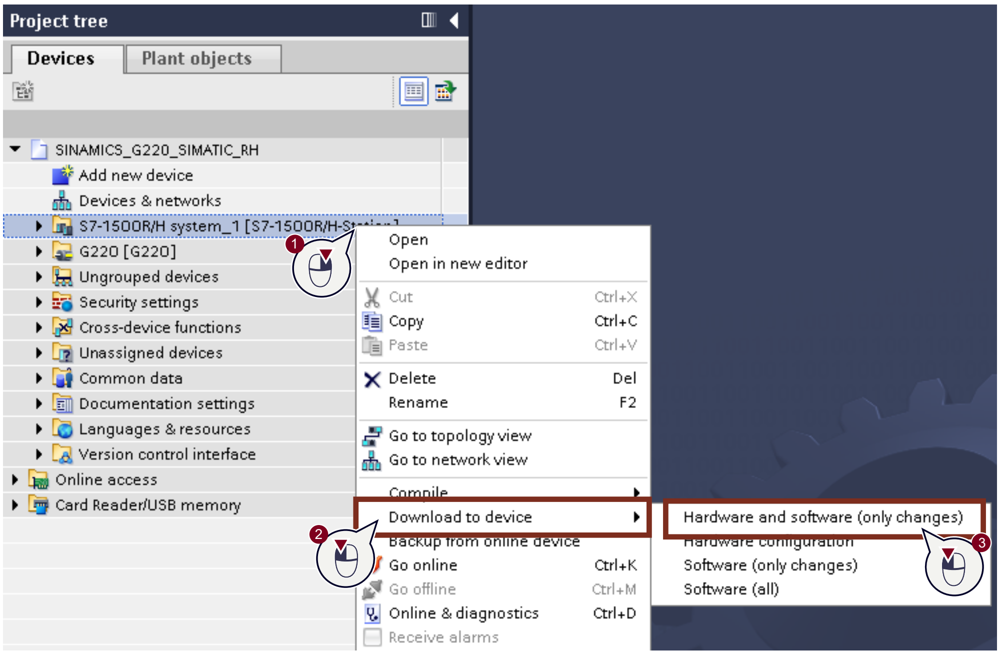
- As a configured GSDML file for connection to and communication with the redundant system
- As an engineered drive in Startdrive without a communication connection, only for parameterization of the drive.

If you are using other hardware and have configured it as described in this document, you can also commission your own project by following the steps below:

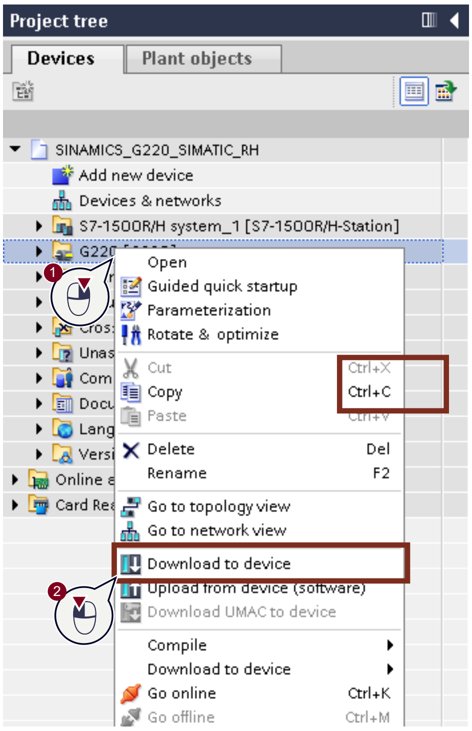
No.	Action
1.	Unzip and open the sample project in TIA Portal.



No.	Action
2.	Download the configuration to the SIMATIC S7 H system by right-clicking on the system in the project tree (1), selecting "Download to device" (2) and then clicking on "Hardware and software (only changes)" (3).

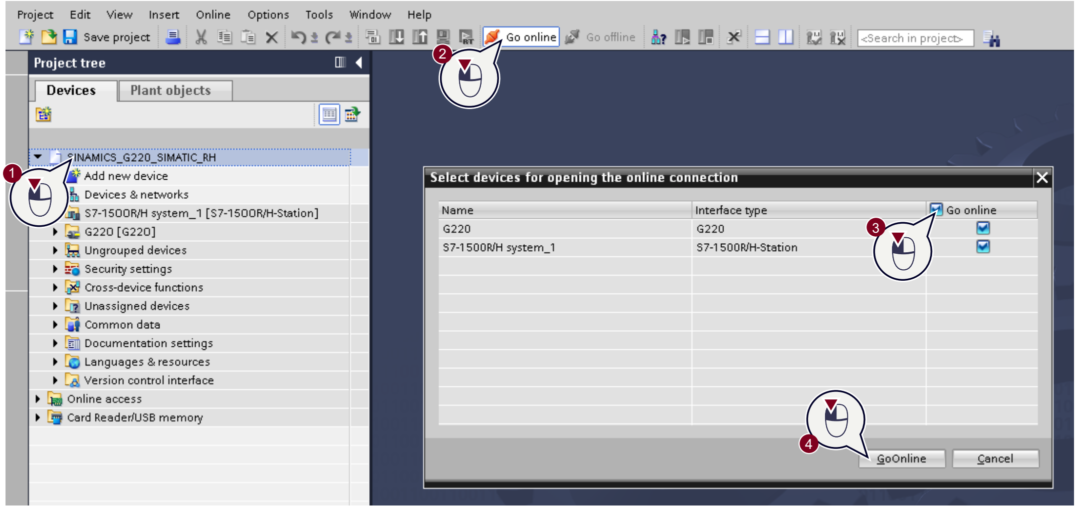


No.	Action
3.	Download the configuration to the SINAMICS G220 drive by right-clicking to select the drive in the project tree (1) and then clicking on "Download to device" (2). **Note:** In the example project, user management for the SINAMICS G220 is enabled. Therefore, you must use the following access data for the download: User name: Siemens Password: Siemens123



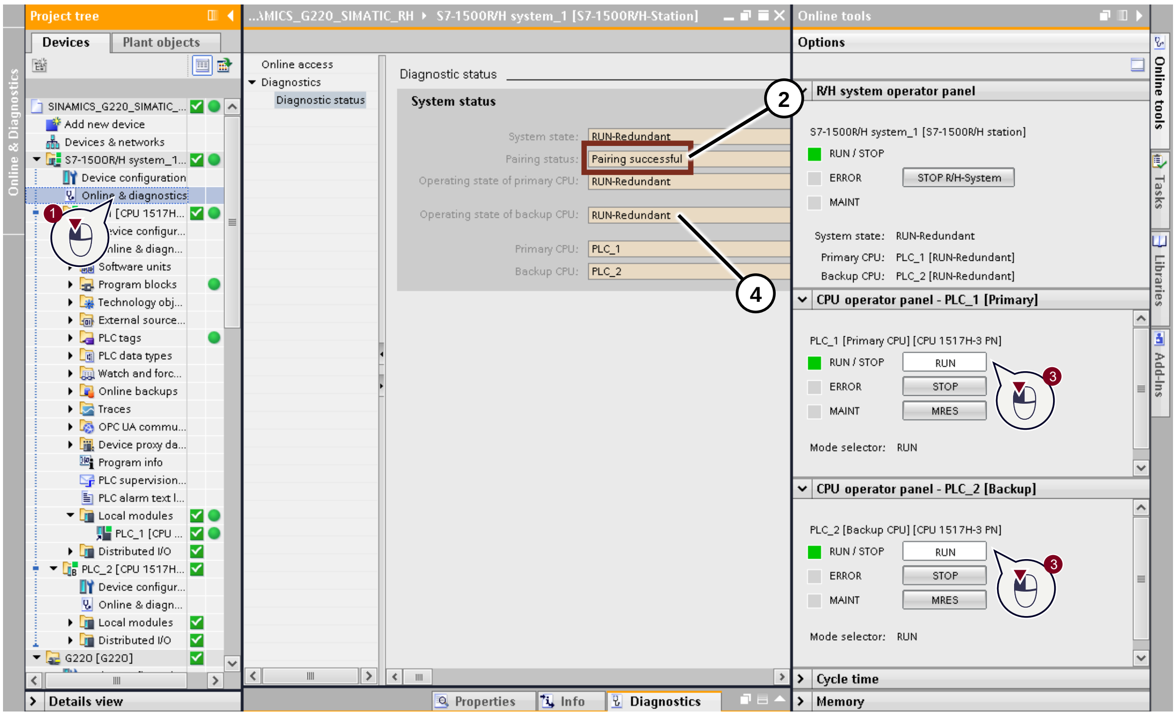
No. **Action**

4. Go online with all devices by selecting the project in the project tree (1), then clicking on "Go online", selecting all devices in the dialog (3) and clicking on "Go online".

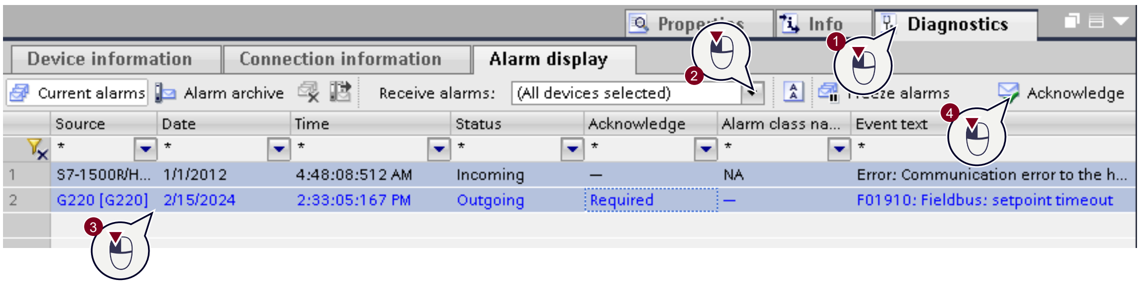


5. Open the diagnostics of the S7-1500 fault-tolerant system (1) and check that both controllers are paired (2). Once the controllers are paired, set both controllers to "RUN" mode (3).

Note:
You may need to go online again after switching to "RUN" mode. Once the controllers are both in "RUN" mode, this will be indicated in the diagnostics (4)



6. Acknowledge any errors that may have occurred by opening the diagnostics in the Inspector window (1), selecting all devices (2), selecting the errors in the message pane (3) and clicking on "Acknowledge" (4).



3.2. Principle of operation

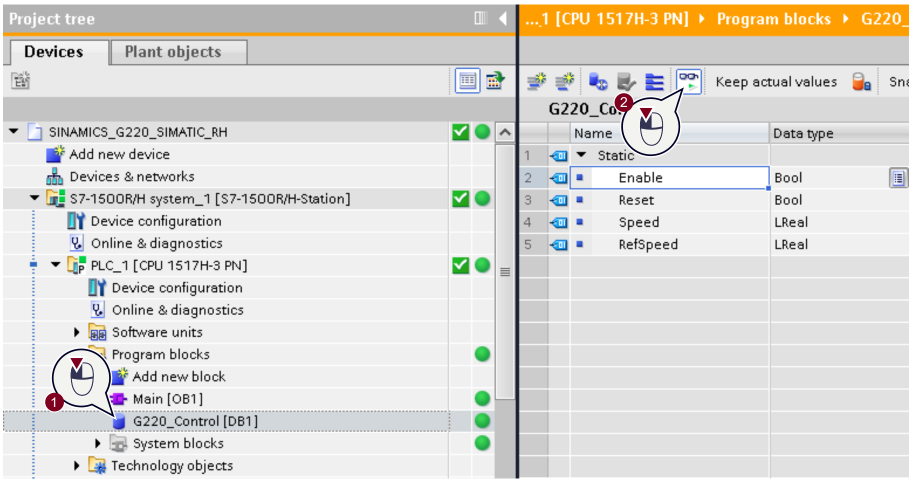
The example project uses the "SinaSpeed" block to control the drive. To control the block, the "G220_Control" data block has been created in the project. It contains the necessary signals for controlling the drive.

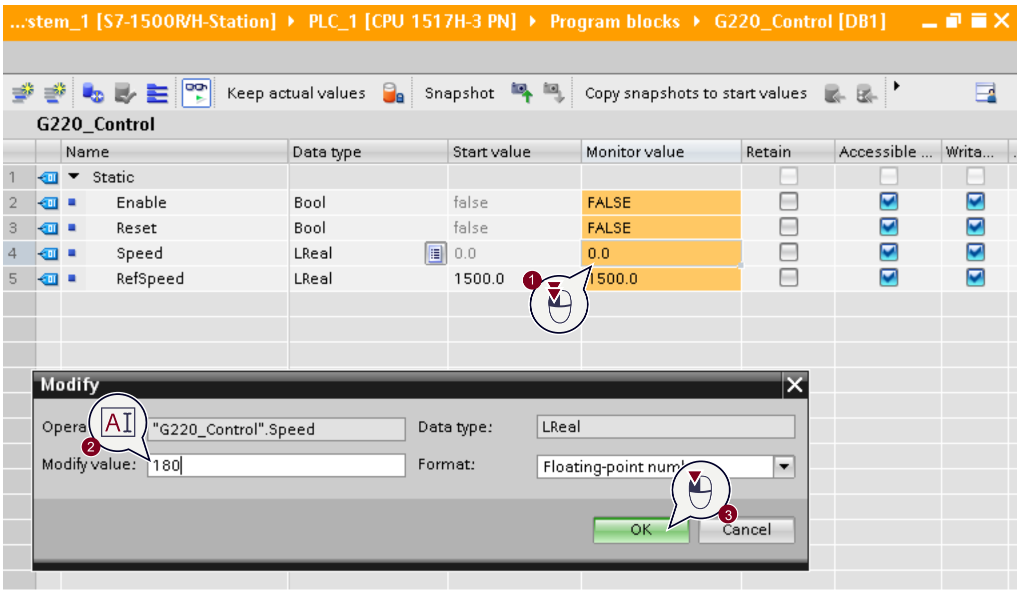
NOTE

For more information on the use of the SINAMICS G220 together with SIMATIC S7 controllers and controlling a drive via the SinaSpeed block, please refer to the application example "Speed control of a SINAMICS G220 with SIMATIC S7-1500 via PROFINET" (\5\)

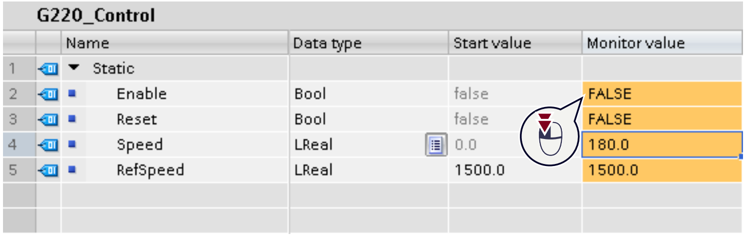
When the drive is moving, you can test the S2 redundancy, for example, by turning off the power supply to the primary controller in the redundant system. This process is described in the following table:

No.	Action
1.	Open the "G220_Control" data block and observe the current values by clicking on "Monitor all".
2.	Double-click on "Speed" (1) and specify a desired setpoint speed (e.g. 180 rpm) (2). Confirm the setpoint speed with "OK" (3).





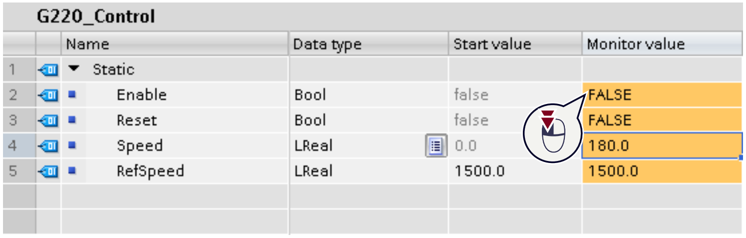
No.	Action
3.	Double-click on "Enable" to start the drive at the specified speed.



G220_Control

	Name	Data type	Start value	Monitor value
1	Static			
2	Enable	Bool	false	FALSE
3	Reset	Bool	false	FALSE
4	Speed	LReal	0.0	180.0
5	RefSpeed	LReal	1500.0	1500.0

No.	Action
4.	Turn off the power supply to the primary CPU while the drive is moving.
5.	The drive continues to operate. The former backup CPU is now the primary CPU. Switch the power supply back on. The former primary CPU boots up again as a backup CPU.
6.	Go online again on the SIMATIC H system and stop the drive by double-clicking again on the current value of "Enable".



G220_Control

	Name	Data type	Start value	Monitor value
1	Static			
2	Enable	Bool	false	FALSE
3	Reset	Bool	false	FALSE
4	Speed	LReal	0.0	180.0
5	RefSpeed	LReal	1500.0	1500.0

4. Appendix

4.1. Service and support

SiePortal

The integrated platform for product selection, purchasing and support - and connection of Industry Mall and Online support. The SiePortal home page replaces the previous home pages of the Industry Mall and the Online Support Portal (SIOS) and combines them.

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- mySieportal
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For more information on our offered trainings and courses, as well as their locations and dates, refer to our web page: siemens.com/sitrain

Industry Online Support app

You will receive optimum support wherever you are with the "Industry Online Support" app. The app is available for iOS and Android:

4.2. Links and literature

No.	Topic
\1\	Siemens Industry Online Support https://support.industry.siemens.com
\2\	Link to the article page of the application example https://support.industry.siemens.com/cs/ww/en/view/109926733
\3\	Manual: SIMATIC S7-1500 S7-1500R/H redundant system https://support.industry.siemens.com/cs/ww/en/view/109754833
\4\	Manual: SIMATIC S7-1500/ET 200MP Automation system https://support.industry.siemens.com/cs/ww/en/view/59191792
\5\	Application example: "Speed control of a SINAMICS G220 with SIMATIC S7-1500 via PROFINET" https://support.industry.siemens.com/cs/ww/en/view/109955055

4.3. Change documentation

Version	Date	Change
V1.0	03/2024	First version

